Sahaj Bhadja

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Summary

Software engineer with 2+ years developing **data processing workflows** and backend systems using **Python**. Experienced with **ETL pipelines**, **SQL** and relational databases, APIs, and **distributed systems**. Strong background in system integrations, performance optimization, and collaborative problem-solving.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Database Systems, Distributed Systems, Analysis of Algorithms, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, Database Management, Software Engineering, Data Processing

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and developed scalable **data processing workflows** using **Python**, building backend components for **data ingestion, transformation, and storage** handling 5K+ monthly requests.
- Implemented **system integrations using APIs** and data synchronization mechanisms, monitoring system performance and participating in troubleshooting and **root cause analysis**.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **ETL pipelines** and data processing workflows using **Python**, optimizing database structures and queries for improved performance in **large-scale data systems**.
- Deployed data services in **containerized environments** using Docker, collaborating with cross-functional teams to gather requirements and deliver technical solutions.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed **data processing pipelines** for ML workflows using **Python and SQL**, supporting **large-scale distributed data systems** and improving processing efficiency by 18%.
- Optimized **database structures and data models** in PostgreSQL, documenting system designs, workflows, and technical processes for knowledge transfer.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Designed and maintained backend components for **data ingestion and transformation**, implementing **RESTful APIs** and webhooks for system integrations.
- Supported testing, debugging, and **continuous improvement** of data-driven applications, using **Git** for version control in collaborative development environment.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built integration solutions using **APIs and data synchronization**, assisting with deployment and configuration of data services in structured development environments.
- Collaborated with cross-functional teams to gather requirements, demonstrating strong **problem-solving skills** and ability to work in collaborative environment.

Skills

Languages: Python, Java, SQL, JavaScript, Bash
Data Engineering: ETL pipelines, data processing workflows, data ingestion/transformation
Databases: PostgreSQL, MySQL, MongoDB, query optimization, data modeling
Tools & DevOps: Git, Docker, CI/CD, APIs, webhooks, system monitoring, AWS

Projects

DataFlow Pipeline | *Python, PostgreSQL, Docker, ETL*

- Built scalable **ETL pipeline** for data ingestion and transformation, processing 100K+ records daily with optimized SQL queries and automated data validation workflows.

Big-Brain Live | *Python, PostgreSQL, APIs, Data Processing*

- Developed **data processing workflows** for document intelligence platform, implementing API integrations and database operations for large-scale data management.

Chat2SQL Code | *Python, MySQL, SQL Optimization, APIs*

- Created **data-driven application** converting natural language to optimized SQL queries, demonstrating strong SQL skills and database integration capabilities.